Mark schemes

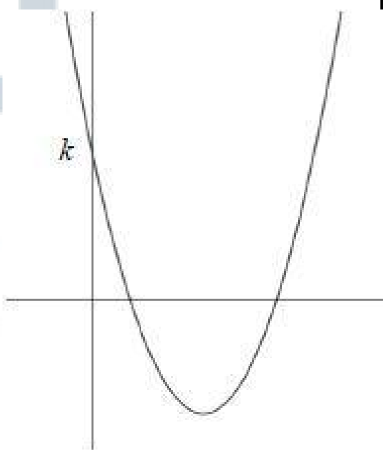
Q1.

Marking Instructions	AO	Marks	Typical Solution
Ticks correct answer	AO1.1b	B1	Figure 3
			Total 1 mark

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$y = 5 \times 5^x$
			Total 1 mark

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Sketches graph recalling correct u shape	AO1.2	B1	
	Deduces correct relative positions of intersections with axes and with k labelled	AO2.2a	B1	
(b)	Shows evidence of discriminant being used or completing the square to find vertex	AO1.1a	M1	For distinct roots $b^2 - 4ac > 0$ $(-6)^2 - 4 \times 1 \times k > 0$ $36 - 4k > 0$ $k < 9$
	Obtains $k < 9$ Condone $k \leq 9$	AO1.1b	A1	
	Explains that positive roots and the u shape of the graph (OE) mean the graph must cross the y-axis above 0 or $k > 0$.	AO2.4	E1	k is the y-intercept and for positive roots the intercept must be positive

States correct range of values for k	AO2.2a	R1	$0 < k < 9$
			Total 6 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Compares with $R\cos(x \pm \alpha)$ or $R\sin(x \pm \alpha)$	AO3.1a	M1	$\sqrt{3}\sin x - 3\cos x \equiv R\sin(x-\alpha)$ $\equiv R\sin x\cos\alpha - R\cos x\sin\alpha$ $R\cos\alpha = \sqrt{3}$ $R\sin\alpha = 3$ $R = \sqrt{12} = 2\sqrt{3}$ $\tan\alpha = \sqrt{3}$ $\alpha = \frac{\pi}{3}$ $y = 2\sqrt{3}\sin\left(x - \frac{\pi}{3}\right) + 4$ Translation $\begin{pmatrix} \frac{\pi}{3} \\ 0 \end{pmatrix}$ Stretch in the y-direction scale factor $2\sqrt{3}$ Translation $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$
	Obtains two correct equations for R and α for example $R\cos\alpha = \sqrt{3}$ $R\sin\alpha = 3$ Must be explicitly seen	AO3.1a	M1	
	Obtains correct R Condone AWRT 3.46 PI by description of stretch	AO1.1b	B1	
	Obtains correct α in radians or degrees PI by description of translation	AO1.1b	B1	
	Interprets their values of R and α to form an equation of the form $y = R\sin(x \pm \alpha) + 4$ or $y = R\cos(x \pm \alpha) + 4$	AO3.2a	B1F	
	Interprets 'their' equation to identify a transformation	AO3.2a	E1F	
	Identifies all required transformations in a correct order CAO	AO3.2a	A1	
(b)(i)	Deduces the least value occurs when their $\sin\left(x - \frac{\pi}{3}\right) = 1$ Using 'their' values of R and α PI by sight of $\frac{1}{2\sqrt{3}+4}$	AO2.2a	M1	$\frac{1}{\sqrt{3}\sin x - 3\cos x + 4} = \frac{1}{2\sqrt{3}\sin\left(x - \frac{\pi}{3}\right) + 4}$ Least value when $\sin\left(x - \frac{\pi}{3}\right) = 1$ $\therefore$ least value is given by $\frac{1}{2\sqrt{3}+4} = \frac{2-\sqrt{3}}{2}$

Completes rigorous argument to obtain $\dfrac{1}{2\sqrt{3}+4}$ and then the given answer	AO2.1	R1	

(ii)	Deduces the greatest value using 'their' values of R and α ACF $\dfrac{1}{-2\sqrt{3}+4}=\dfrac{2+\sqrt{3}}{2}$	AO2.2a	B1F	Greatest value $=\dfrac{2+\sqrt{3}}{2}$
				Total 10 marks

Q5.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$\begin{bmatrix} -4 \\ 0 \end{bmatrix}$
			Total 1 mark

Q6.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.2	B1	$y = \mathrm{f}(2x)$
			Total 1 mark

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Compares with $R\cos(\theta \pm \alpha)$ or $R\sin(\theta \pm \alpha)$	AO3.1a	M1	$R\cos(\theta - \alpha)$ $\equiv R\cos\alpha\cos\theta + R\sin\theta\sin\alpha$ $\therefore R\cos\alpha = 3$ and $R\sin\alpha = 3$ $R = \sqrt{18}$
	Identifies version which will allow them to solve the problem	AO3.1a	A1	
	Obtains correct R	AO1.1b	A1	

	Obtains correct α	AO1.1b	A1	$\alpha = \frac{\pi}{4}$
	Interprets 'their' equation to identify first transformation	AO3.2a	E1	$\therefore 3\cos\theta + 3\sin\theta \equiv \sqrt{18}\cos\left(\theta - \frac{\pi}{4}\right)$
	Interprets 'their' equation to identify second transformation	AO3.2a	E1	Which is a stretch in the y-direction scale factor $\sqrt{18}$ and a translation $\begin{pmatrix} \frac{\pi}{4} \\ 0 \end{pmatrix}$
(b)	Constructs a rigorous mathematical argument, to find either the least or greatest value Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips (no FT for this mark)	AO2.1	R1	4 + (3cos θ + 3sin θ)^2 $4 + \left(\sqrt{18}\cos\left(\theta + \frac{\pi}{4}\right)\right)^2$ Least value occurs when $\cos^2\left(\theta + \frac{\pi}{4}\right) = 0$ ∴least value = 4
	Deduces the least value Using 'their' values of R and α	AO2.2a	A1F	Greatest value occurs when $\cos^2\left(\theta + \frac{\pi}{4}\right) = 1$
	Deduces the greatest value Using 'their' values of R and α	AO2.2a	A1F	Greatest value = 4 + 18 = 22
				Total 9 marks

Q8.

(a) (i) $(x - 3)^2$

or p = 3 seen

M1

$(x - 3)^2 + 2$

A1

2

(ii) $(x - 3)^2 = -2$

FT their positive value of q

not *use of discriminant*

M1

No (real) square root of −2 therefore equation has no real solutions

for graphical approach see below to see if SC1 can be awarded

A1cso

2

(b) (i) x = 'their' p *or* y = 'their' q

or x = 3 found using calculus

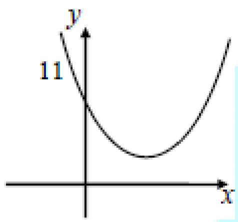
M1

Vertex is at (3, 2)

A1cao

2

(ii)



y intercept = 11 ***stated*** *or* ***marked on y-***axis *(as y intercept of any graph)*

B1

∪ shape (generous)

M1

above x-axis , vertex in first quadrant crossing y-axis into second quadrant

A1

3

(iii) Translation

and no other transformation

E1

through $\begin{bmatrix} -3 \\ -2 \end{bmatrix}$

FT negative of BOTH 'their' vertex coords

M1

both *components* ***correct*** *for A1; may describe in words or use a column vector*

A1

3

[12]

Q9.

(a) Stretch(**I**) in y-direction(**II**) scale factor 3(**III**)

*OE Need (**I**) **and** either (**II**) or (**III**)*

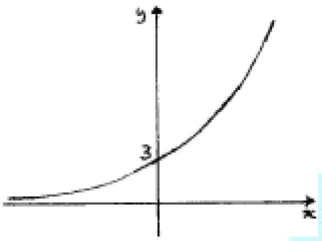
M1

*All correct. Need (**I**) and (**II**) and (**III**)*
[>1 transformation scores 0 / 2]

A1

2

(b)



Shape with indication of correct asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis, and only intercept is 3 stated / indicated

B1

2

(c) $3 \times 4^x = 4^{-x}$

OE eqn. in x

M1

$\log 3 + \log 4^x = \log 4^{-x}$

Log Law 1 (or Law 2 applied to $\frac{4^x}{4^{-x}} = 3$ or $\frac{1}{3}$ OE) used correctly or correct rearrangement to $4^{2x} = 1/3$ OE simplified e.g. $16^x = 3^{-1}$ or $4^x = (1/\sqrt{3})$

m1

$\log 3 + x\log 4 = -x\log 4$

Log Law 3 applied correctly twice (dependent on both M1 & m1) or a correct method using logs to solve an eqn. of form $a^{kx} = b$, $b > 0$ (including case $k = 1$) (dependent *on M1 and valid method to a^{kx})*

m1

$x = \dfrac{-\log 3}{2\log 4} \quad \left(= \dfrac{-\log 3}{\log 16}\right)$

Correct expression for x or for $-x$

e.g. $x = \frac{1}{2}\log_4\left(\frac{1}{3}\right)$

PI by correct 3sf value or better

A1

$x = -0.396(2406\ \ldots) \quad = -0.396$ (to 3sf)

If logs not used explicitly then max of M1m1m0.

A1

5

[9]

Q10.

(a) $x + 30° = 79°$, $x + 30° = 180° + 79°$

$x = 49°$

49 as the only solution in the interval $0° \le x \le 90°$

B1

$x = 229°$

AWRT 229. Not given if any other soln. in the interval $90° \le x \le 360°$.

Ignore anything outside $0° \le x \le 360°$

B1

2

(b) Translation;

Accept 'translat…' as equivalent.
[T or Tr is NOT sufficient]

B1

$\begin{bmatrix} -30° \\ 0 \end{bmatrix}$

*OE Accept **full** equivalent to vector in words provided linked to 'translation / move / shift' and **correct** direction. (0 / 2 if >1 transformation).*

B1

2

(c) (i) $5 + \sin^2\theta = (5 + 3\cos\theta)\cos\theta$

$\Rightarrow 5 + \sin^2\theta = 5\cos\theta + 3\cos^2\theta$

Correct RHS.

B1

$5 + 1 - \cos^2\theta = 5\cos\theta + 3\cos^2\theta$

$\sin^2\theta = 1 - \cos^2\theta$ used to get a quadratic in cos.

M1

$6 = 5\cos\theta + 4\cos^2\theta$ or $4\cos^2\theta + 5\cos\theta - 6$ $(= 0)$

ACF with like terms collected.

A1

$\Rightarrow$ $(4\cos\theta - 3)(\cos\theta + 2)$ $(= 0)$

Correct *quadratic and $(4c \pm 3)(c \pm 2)$ or by formula OE PI by $\pm$ 'correct' 2 values for cos θ.*

m1

Since $\cos\theta \neq -2$, $\cos\theta = \frac{3}{4}$

CSO AG. Must show that the 'soln' $\cos\theta = -2$ has been considered and rejected

A1

5

(ii) $5 + \sin^2 2x = (5 + 3\cos 2x)\cos 2x \Rightarrow \cos 2x = \frac{3}{4}$

Using (c)(i) to reach cos2x = ¾ ***or*** *finding at least 3 solutions of cos θ = ¾* ***and*** *dividing them by 2.*

M1

$2x = 0.722(7...)$, $2\pi - 0.722(7...)$,
$2\pi + 0.722(7...)$, $4\pi - 0.722(7...)$,

Valid method to find all four 'positions' of solutions.

m1

$x = 0.361$, 2.78 , 3.50, 5.92

CAO Must be these four 3sf values but ignore any values outside the interval $0 < x < 2\pi$.

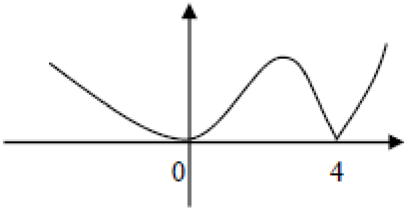
A1

3

[12]

Q11.

(a)



reflection in the x-axis for the negative f(x) and remainder as given on sketch

M1

correct curvatures, correct cusp at x = 4

condone straight lines for x 4
4 marked on x-axis

A1

2

(b) **Either**
1. Stretch

1 and either 2 or 3

M1

2. || x-axis

3. by factor 0.5

1, 2 and 3

A1

(followed by) translation

E1

$$\begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$$

B1

4

or
translation

(E1)

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(B1)

(followed by) 1. Stretch

1 and either 2 or 3

(M1)

2. || x-axis

3. by factor 0.5

1, 2 and 3

(A1)

(4)

[6]

Q12.

(a) $(x - 5)^2 + (y + 7)^2$

one term correct

M1

both terms correct and added

A1

$(x - 5)^2 + (y + 7)^2 = 49$

must see 49 not just 7^2

condone $(x - 5)^2 + (y - -7)^2 = 49$

A1cao

3

(b) (i) (Centre is) (5, −7)

correct or FT their a *and* b

B1√

1

(ii) Radius = 7

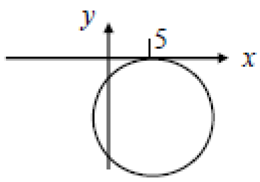
*condone 49 but **not*** ± 7 *or* $\pm\sqrt{49}$

correct or FT their $\sqrt{k}$ *provided* $k > 0$

B1√

1

(c) (i)



freehand circle with centre in correct quadrant or FT from their (b)(i) must have both axes shown clearly

M1

correct position cutting negative y-axis twice and touching x-axis at x = 5
5 must be marked on x-axis or centre clearly marked as (5, −7)
must have correct centre and radius in (b)

A1

2

(ii) $x = 5$

B1

$y = -14$

(5, −14)

B1

2

(d) Translation

and no other transformation

E1

through $\begin{bmatrix} 6 \\ * \end{bmatrix}$

M1

$\begin{bmatrix} 6 \\ -7 \end{bmatrix}$

both components correct for A1; may describe in words or use a column vector

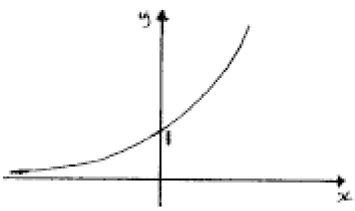
A1cso

3

[12]

Q13.

(a)



Correct graph, must clearly go below the intersection pt

and an indication of correct behaviour of curve for large positive and large negative values of x.
Ignore any scaling on axes.

B1

Only one y−intercept, marked / stated as 1 or as coords (0, 1) with graph having no other intercepts on either axes.

B1

2

(b) $9^x = 15 \Rightarrow x \log 9 = \log 15$

OE eg $x = \log_9 15$

M1

(x =) 1.23(2486…) = 1.23 to 3sf

Condone > 3sf.
Must see evidence of logs used so NMS scores 0/2

A1

2

(c) {f(x) =} 9^{-x}

OE

B1

1

[5]

Q14.

(a) $h = 0.5$

h = 0.5 stated or used.

B1

f(x) = $\sqrt{8x^3+1}$

$I \approx \frac{h}{2}$ {f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]}

$I \approx \frac{h}{2}$ {f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]} OE

M1

$\frac{h}{2}$ with {…} = $\sqrt{1}+\sqrt{65}+2\left(\sqrt{2}+\sqrt{9}+\sqrt{28}\right)$

= 1 + 8.06... + 2(1.41… + 3 + 5.29…)

= 9.0622… + 2 × 9.7057…

OE Accept 1dp evidence. Can be implied by later correct work provided more than one term or a single term which rounds to 7.12

A1

(I ≈) 0.25[28.47…] {= 7.118..} = 7.12 (to 3sf)

CAO Must be 7.12

A1

(b) Stretch(**I**) in x−direction(**II**) scale factor 2 (**III**)

*Need (**I**) and either (**II**) or (**III**)*

M1

*Need (**I**) and (**II**) and (**III**)*

More than 1 transformation scores 0/2

A1

(c) g(x) = $\sqrt{(x-2)^3+1}-0.7$

$\sqrt{(x-2)^3+1-0.7}$ *or* $\sqrt{(x-2)^3+1}+0.7$

or $\sqrt{(x+2)^3+1}-0.7$ *or* $\sqrt{(x-2)^3+1}-0.7$

or their equivalents

M1

$\sqrt{(x-2)^3+1}-0.7$ *OE*

A1

g(4) = 2.3

2.3 OE

A1

3

Alternative:

(4, …) on y = g(x) comes from translating (2, 3) on y = $\sqrt{x^3+1}$

from (2, …) on y = $\sqrt{x^3+1}$

(M1)

from (2, 3) on y = $\sqrt{x^3+1}$

(A1)

(2, 3) after translation becomes (4, 2.3) so g(4) = 2.3

2.3 OE

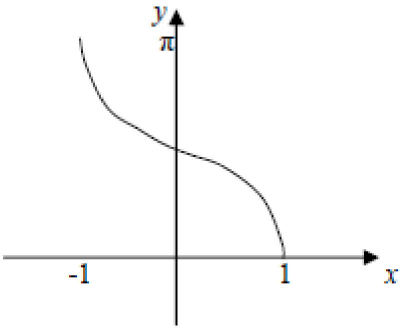
(A1)

(3)

[9]

Q15.

(a)



Correct sketch of cos^{-1} x.

B1

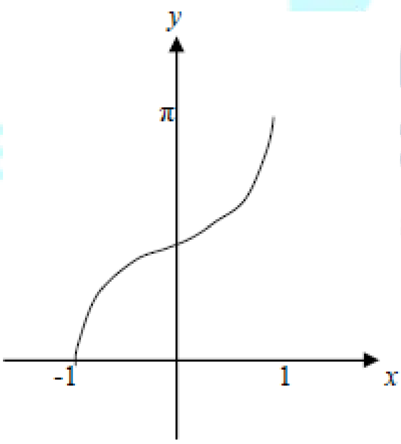
(−1, π) and (1, 0)

Stated

B1

2

(b)



Correct sketch of π − *cos*$^{-1}$ x

Must touch negative x*-axis.*

B1

(−1, 0) and (1, π)

Stated

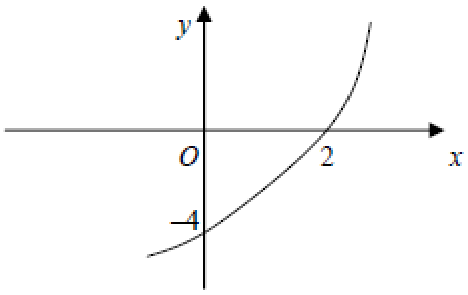
B1

2

[4]

Q16.

(a)



Reflection in the x-axis.

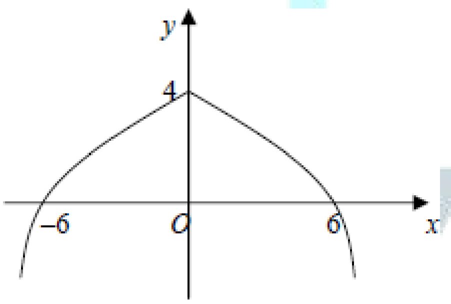
M1

Intersection with the x-axis and y-axis marked 2 and −4.
Accept (2, 0) and (0, −4) instead of marking on the axes.

A1

2

(b)



Reflection of $0 < x < 6$ part in the y-axis giving two connected sections

M1

Correct curve beyond ± 6, correct curvature and correct cusp at x = 0 (generous)

A1

±6 and 4 marked correctly
Accept (6, 0), (0, 4) and (−6,0) instead of marking on the axes.

A1

3

(c) Reflection

M1

in the y−axis

A1

(followed by)

1) stretch
2) parallel to the x−axis
3) by factor 2

1 and either 2 or 3

M1

1, 2 and 3

OR
Vice versa

A1

4

[9]

Q17.

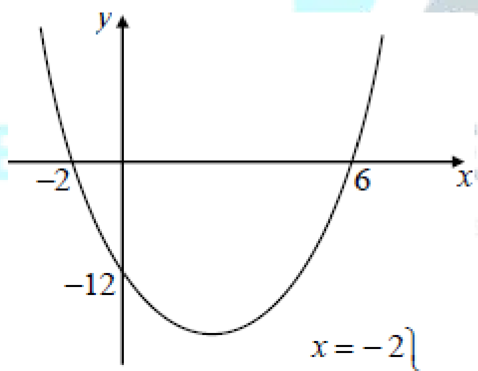
(a) $(x - 6)(x + 2)$

ISW for $x = 6$, $x = -2$ etc

B1

1

(b)



$$\left.\begin{matrix} x = -2 \\ x = 6 \end{matrix}\right\}$$

correct x values or FT 'their' factors (x-intercepts stated or marked on sketch) may be seen in (a)

B1√

$y = -12$

(stated or −12 marked on sketch)

B1

∪ − shaped curve

approximately

M1

"correct" shape in all 4 quadrants with minimum to right of y-axis

A1

(c) (i) $(x - 2)^2$

$p = 2$

M1

$(x - 2)^2 - 16$

$p = 2$ and $q = 16$

A1

(ii) (Minimum value is) -16

FT ' their $-q$ '

B1√

(d) Replacing each x by size:125%;font-family:Times New Roman;'>x + 3 **OR** adding 2 to their quadratic

in original equation or 'their' completed square or factorised form or replacing y by $y - 2$

M1

$y = [(x + 3)^2 - 4(x + 3) - 12] + 2$

or $y = (x + 1)^2 - 14$

or $y = x^2 + 2x - 13$

or $y - 2 = (x - 3)(x - 5)$

OE any correct equation in x and y ***unsimplified***

A1 2

[10]

Q18.

(a) (i) Stretch(**I**) in x-direction(**II**)

*Need (**I**) and either (**II**) or (**III**)*

M1

scale factor $\frac{1}{6}$ (**III**)

*Need (**I**) and (**II**) and (**III**)*

A1

2

(ii) (g(x) =) = $\left(1+\frac{x-3}{3}\right)^6$

OE Replaces $\frac{x}{3}$ by $\frac{x-3}{3}$

M1

= $\left(\frac{x}{3}\right)^6$ or $\frac{x^6}{3^6}$ or $\frac{x^6}{729}$

Must be simplified

A1

2

(b) $\left(1+\frac{x}{3}\right)^6 = 1+\binom{6}{1}\frac{x}{3}+\binom{6}{2}\left(\frac{x}{3}\right)^2+\binom{6}{3}\left(\frac{x}{3}\right)^3$ …

= (1 +) 2x

a = 2. Condone '2x'

B1

$+\frac{6!}{4!2!}\left(\frac{x}{3}\right)^2+\frac{6!}{3!3!}\left(\frac{x}{3}\right)^3$

Either (1 6) 15 20 seen or $\binom{6}{2}$, $\binom{6}{2}$ *written (PI) in terms of factorials (OE)*

M1

= (1 + 2x) $+\frac{15}{9}x^2+\frac{20}{27}x^3$

(a = 2)

$b = \frac{5}{3}$, $c = \frac{20}{27}$

$b = \frac{5}{3}$ (or 1 $\frac{2}{3}$). Condone… + $\frac{5}{3}x^2$

A1

$c = \frac{20}{27}$. Condone…$+\ \frac{20}{27}x^3$

A1

Accept equivalent recurring decimals Ignore terms with higher powers of x **SC** *If $A0A0$ award $A1$ for either*

$+15\frac{x^2}{9},\ +20\frac{x^3}{27}$ *seen or* $+\frac{15x^2}{9},\ +\frac{20x^3}{27}$ *seen*

4

[8]

Q19.

(a)

stretch I

SF 4 II

in y-direction III

translate $\begin{pmatrix} e \\ 0 \end{pmatrix}$

either order

I + (II or III)

M1A1

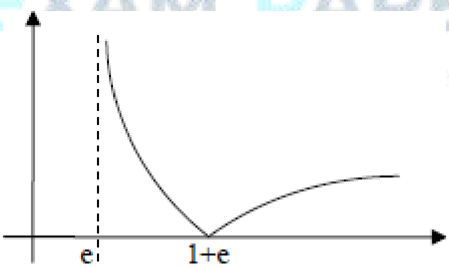
E1

accept 'e in positive x-direction'

B1

4

(b)



mod graph, in 2 connected sections, both in the first quadrant, touching x-axis

M1

curve touches x-axis at 1 + e (or 3.7 or better), and labelled (ignore scale)

A1

correct curvature, including at their 1 + e, approx. asymptote at $x = e$

A1

3

(c) (i) $| 4\ln(x - e) | = 4$

$4\ln(x - e) = 4$ $4\ln(x - e) = -4$ or better

must see 2 equations, condone omission of brackets

M1

$(x =)\ 2e$ do not ISW

accept values of AWRT 5.42, 5.43, 5.44

A1

$(x =)\ e + e^{-1}$ or $(x =)\ e + \frac{1}{e}$ do not ISW

accept values of AWRT 3.08, 3.09

if M0 then x = 2e with or without working scores SC1

A1

3

(ii) $x \geq 2e$

accept values of AWRT 5.42, 5.43, 5.44

B1

$e < x \leq e + \frac{1}{e}$

accept values of AWRT 2.72, 3.08, 3.09

if B2 not earned, then SC1for any of

$e \leq x \leq e + \frac{1}{e}$, $e < x < e + \frac{1}{e}$, $e \leq x < e + \frac{1}{e}$

B2

3

[13]

Q20.

(a) $h = 0.25$

PI

B1

$f(x) = \log_{10}(x^2 + 1)$

$I \approx h/2\{\ldots\}$

$\{.\} = f(0) + f(1) + 2[f(0.25) + f(0.5) + f(0.75)]$

OE summing of areas of the 'trapezia'

M1

$\{.\} = \log 1 + \log 2 + 2\left[\log\frac{17}{16} + \log\frac{5}{4} + \log\frac{25}{16}\right]$

OE Accept 1sf evidence

A1

= 0 + 0.3010... + 2(0.0263... + 0.0969... + 0.1938...)

= 0.3010...+ 2(0.317058...) = 0.935147 …

(I ≈) 0.125 [0.935147 …] = 0.117 (to 3SF)

CAO Must be 0.117

A1

4

(b) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

B1

1

(c) (i) $\log_{10}(10x^2) = \log_{10}10 + \log_{10} x^2$

Condone missing bases for M mark.

Accept log x^2 replaced by 2log x in M1 line

M1

$= 1 + 2\log_{10} x$

AG. Bases must be included or statement '$\log_{10} 10 = 1$' given.

A1

Condone missing bases in (c)(ii) & (c)(iii)

2

(ii) $y = 1 + 2\log_{10} x = \log_{10}(10x^2)$

PI

M1

Either $y = 2\log_{10}(\sqrt{10}x)$ (to compare $y = 2\log x$)

or both $y = \log_{10} x^2$ and $y = \log_{10}(\sqrt{10}x)^2$

Writing in correct form so that stretch details can be stated directly

A1

(Stretch) parallel to x-axis, sf $\frac{1}{\sqrt{10}}$ OE

B2 for correct direction and scale factor ACF
(B1 for correct exact scale factor ACF)
(or B1 for 'x-direction, scale factor 1/10')
(or B1 for 'x-direction, scale factor $\sqrt{10}$')
Apply ISW if dec follows exact values.
(OE scale factor must be in exact form)

B2,1,0

4

(iii) $\log_{10}(10x^2) = \log_{10}(x^2 + 1)$

PI by $10x^2 = x^2 + 1$ or correct x

M1

$(10x^2 = x^2 + 1,\ 9x^2 = 1$ and since $x > 0)$ $x = \frac{1}{3}$

$x = \frac{1}{3}$ *OE stated or used; accept* $\sqrt{\frac{1}{9}}, \frac{1}{\sqrt{9}}$

A1

(y−coordinate of P) $y = \log_{10}\frac{10}{9}$

Or $y = \log\left(\frac{1}{9}+1\right)$

PI by 3 log $\frac{10}{9}$ OE for the gradient of OP

A1

Gradient of $OP = 3\log_{10}\frac{10}{9} = \log_{10}\frac{1000}{729}$

log $\frac{1000}{729}$; Accept 'a = 1000, b = 729'

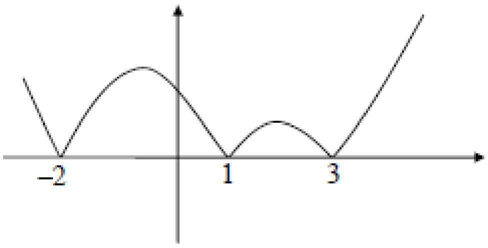
A1

4

[15]

Q21.

(a)



Modulus graph, 4 sections touching x-axis at − 2, 1, 3

M1

Correct $x > 3$, $x < -2$

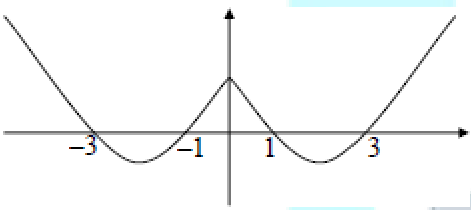
A1

Correct $-2 \le x \le 3$ with maximum at 2 lower than maximum at − 1and correct cusps at $x = -2$, $x = 1$ and $x = 3$
The maximums need to be at $x = -1$ and 2 (approx)

A1

3

(b)



Symmetrical about y-axis, from original curve for $0 < x < 1$ and $x > 3$

M1

Correct graph including cusp at $x = 0$

A1

2

(c)

$$\left.\begin{array}{l}\left.\begin{array}{l}\text{Translate}\\ \begin{bmatrix}-1\\0\end{bmatrix}\end{array}\right\}\\ \left.\begin{array}{l}\text{Stretch (I)}\\ \text{sf } \frac{1}{2} \text{ (II)}\\ // \, y\text{-axis (III)}\end{array}\right\}\end{array}\right\} \text{either order}$$

E1
B1

I and (either II or III)

M1

I + II + III

A1

4

(d) $x = -2$

B1

$y = 5$

Each value may be stated or shown as coordinates

B1

2

[11]

Q22.

(a) $h = 0.5$

PI

B1

$f(x) = \sqrt{27x^3 + 4}$

$I \approx h/2\{\ldots\}$

$\{\ldots\} = f(0) + f(1.5) + 2[f(0.5) + f(1)]$

OE summing of areas of the 'trapezia'..

M1

$\{\ldots\} = \sqrt{4} + \sqrt{95.125} + 2(\sqrt{7.375} + \sqrt{31})$

$= 2 + 9.7532... + 2(2.7156\ldots + 5.5677\ldots)$

OE Accept 2dp rounded or truncated as evidence for surds

A1

$(I \approx)$ $0.25 \times 28.32012\ldots = 7.08$ (to 3sf)

Must be 7.08

A1

4

(b) $g(x) = \sqrt{27\left(\frac{1}{3}x\right)^3 + 4} = \sqrt{x^3 + 4}$

Any form which simplifies to $\sqrt{kx^3 + 4}$ *,* $k \neq 27$*,* $k \neq 0$
or which simplifies to $x^3 + 4$

M1

ACF

A1

2

[6]

Q23.

(a) $y = x + 3 + \dfrac{8}{x^4} = x + 3 + 8x^{-4}$

For $\dfrac{8}{x^4} = 8x^{-4}$ *PI by correct differentiation of 3rd term*

B1

$\dfrac{dy}{dx} = 1 - 32x^{-5}$ or $1 - \dfrac{32}{x^5}$

kx^{-5} *OE*

M1

For either

A1

3

(b) When $x = 1$, $y = 12$

B1

When $x = 1$, $\dfrac{dy}{dx} = 1 - 32 = -31$

Attempt to find value of $\dfrac{dy}{dx}$ *when* $x = 1$

M1

Tangent: $y - 12 = -31(x - 1)$
*Only ft on c's answer to (**a**). Any correct (ft on c's (a)) form.*

A1F

3

(c) $1 - 32x^{-5} = 0$

$1 - 32x^{-5} = 0$ or c's $\frac{dy}{dx} = 0$

M1

$\Rightarrow x^5 = 32$

Attempt to form x^n = const (≠0). PI by next line

m1

$\Rightarrow x = 2$

CSO

A1

(Coordinates of *M*) (2, 5.5)

CSO

A1

4

(d) (i) $\int\left(x + 3 + \frac{8}{x^4}\right)dx$

$= \frac{x^2}{2} + 3x - \frac{8}{3}x^{-3} + c$

Power −3 correctly obtained

M1

$-\frac{8}{3}x^{-3}$

A1

$\frac{x^2}{2} + 3x + c$

B1

3

(ii)

$= \left(2 + 6 - \frac{1}{3}\right) - \left(\frac{1}{2} + 3 - \frac{8}{3}\right)$

Attempting to calculate F(2) − F(1) where F(x) is c's answer to part (d)(i) provided F is not just the c's integrand (x + 3 + 8/x^4)

M1

$\frac{9}{2} + \frac{7}{3} = \frac{41}{6}$

OE Accept 6.83 or better provided d(i) used

A1

2

(e) $k = -5.5$

Ft on $-y_M$ from part (c).

B1F

1

[16]

Q24.

(a) $-3 \le f(x) \le 3$

$-3 \le x \le 3$, $-3 < f(x) < 3$

$-3 < f < 3$, $-3 < y < 3$

$-3 \le f < 3$, $-3 < f \le 3$

M1

Allow $-3 \le y \le 3$, $-3 \le f \le 3$

A1

2

(b) (i) $y = 3\cos\frac{1}{2}x$

$\frac{y}{3} = \cos\frac{1}{2}x$

$\cos^{-1}\frac{y}{3} = \left(\frac{1}{2}x\right)$

$x = 2\cos^{-1}\frac{y}{3}$

$y = 2\cos^{-1}\frac{x}{3}$

$\left.\begin{array}{l}\textit{Or } \cos^{-1}\frac{x}{3} = \\ \textit{Swap } x \textit{ and } y\end{array}\right\}$ *Either order*

M1

M1

$$f^{-1}(x) = 2\cos^{-1}\frac{x}{3}$$

A1

3

(ii) $\frac{x}{3} = \cos\frac{1}{2}$

*If incorrect in (b)(i) **BUT** answer in form $p\cos^{-1}(qx)$ (condone p, q = 1)*

*Then $qx = \cos\left(\frac{1}{p}\right)$ M1 **or** $x = f(1)$ M1*

M1

$x = 3\cos\frac{1}{2}$ ISW

$x = 3\cos\frac{1}{2}$ A1

A1

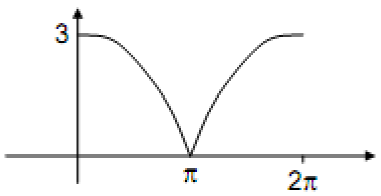
2

(c) (i) $gf(x) = \left|3\cos\frac{1}{2}x\right|$

B1

1

(ii)



Modulus graph in 1st quadrant, starting from a +ve y–intercept, at least 2 continuous parts, first descending, then second increasing
IGNORE CURVE OUTSIDE RANGE

M1

Correct curvature, curves reaching x–axis, condone multiple curves (no turning points at axis)

A1

Approximately symmetrical graph with 3, π, 2π indicated (must have scored previous 2 marks)

Condone $y = 3\cos\frac{1}{2}x$ *also drawn but clearly identified, otherwise M0*

A1

3

(d) STRETCH + direction

Either in x-direction or y-direction

M1

s.f. 3, parallel to y-axis

A1

s.f. 2, parallel to x-axis

Either order

A1

3

[14]

Q25.

(a) $(x + 2.5)^2$

$p = \frac{5}{2}$

B1

$q = 7 - $ ' their' p^2

unsimplified attempt at $q = 7 -$ ' their' p^2

$q = 7 - \frac{25}{4} = \frac{3}{4}$

M1

$(x + 2.5)^2 + 0.75$

mark their final line as their answer

A1

3

(b) (i) $x = -$ 'their' p or $y =$ 'their' q

or $x = -\frac{5}{2}$ cao found using calculus

M1

$\left(-\frac{5}{2}, \frac{3}{4}\right)$

condone correct coordinates stated
$x = -2.5, \quad y = 0.75$

A1cao

2

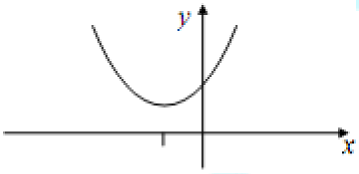
(ii) $x = -\frac{5}{2}$

correct or ft " $x = -$ ' their ' p "

B1 ✓

1

(iii)



y intercept = 7 stated
or seen in table as y = 7 when x = 0
or 7 marked as intercept on y-axis
(any graph)

B1

∪ *shape*

M1

vertex above x-axis in correct quadrant and parabola extending beyond y-axis into first quadrant

A1

3

(c) Translation

and no other transformation

E1

through $\begin{bmatrix} -\frac{5}{2} \\ \frac{3}{4} \end{bmatrix}$

ft either 'their' –p or 'their' q or one component correct for M1

M1

both components correct for A1; may describe in words or use a vector

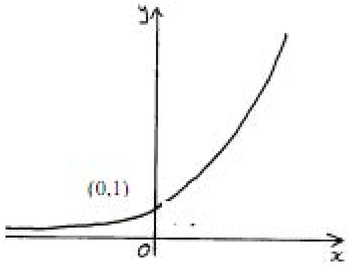
A1cao

3

[12]

Q26.

(a)



*Any graph only crossing the y–axis at (0, 1) stated /indicated … (accept 1 on y–axis as equivalent) and not drawn **below** x–axis*

B1

Correct shaped graph, must clearly go below the intersection pt and an indication of correct behaviour of curve for large positive and large negative values of x. Ignore any scaling on axes.

B1

2

(b) Translation;

Accept 'transl…' as equivalent
[T or Tr is NOT sufficient]

B1

$\begin{bmatrix} 0 \\ -5 \end{bmatrix}$

*If vector not given, accept **full** equivalent to vector in words provided linked to 'transl../move/shift'*
(B0B0 if >1 transformation)

B1

2

(c) (i) $4^x = (2^2)^x = 2^{2x} = (2^x)^2 = Y^2$

$2^{x+2} = 2^x \times 2^2 = 4Y$

Justifying either $4^x = Y^2$ or $2^{x+2} = 4Y$ with no errors seen

M1

$4^x - 2^{x+2} - 5 = 0 \Rightarrow Y^2 - 4Y - 5 = 0$

AG Be convinced; must have justified both of the above.

A1

2

(ii) $(Y - 5)(Y + 1) = 0$

Correct factorising or use of quadratic formula or completing sq. PI by both solns 5 & −1 seen

M1

(Since) $2^x > 0$ (for all real x,) $2^x = 5$ so only one (real) solution

Rejection of 2^x (condone Y) negative, with justification, (condone "2^x not negative") followed by statement

E1

$\log 2^x = \log 5 \Rightarrow x \log 2 = \log 5$

Eqn of form $p^x = q \Rightarrow x\log p = \log q$ provided $p > 0$ & $q > 0$ OE eg $x = \log_2 5$

M1

$x = 2.3219\ldots = 2.322$ (to 3dp)

Condone > 3dp but must see explicit use of logs and must only be the one solution.

A1

4

[10]

Q27.

(a) $h = 0.5$

$h = 0.5$ stated or used. (PI by x–values

0, 0.5, 1, 1.5, 2 provided no contradiction)

B1

$f(x) = \sin x$

$I \approx h/2\{\ldots\}$

$\{.\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$

OE summing of areas of 'trapezia'..

M1

$\{.\} =$

$0 + 0.90929.. + 2[0.4794.. + 0.84147.. + 0.99749..]$

$\{= 0.90929.. + 2[2.318..]\} = \{ 0.90929.. + 4.636..\}$

Min. of 2dp values rounded or truncated. Can be implied by later correct work provided > 1 term or a single term which rounds to 1.39

A1

$(I \approx)\ 0.25[5.546..] = 1.3865.. = 1.39$ (to 3sf)

CAO Must be 1.39

A1

4

(b) Stretch(**I**) in y–direction(**II**) scale factor 2(**III**)

*Need (**I**) and either (**II**) or (**III**)*

M1

*All correct. Need (**I**) and (**II**) and (**III**)*
[> 1 transformation scores 0/2]

A1

2

(c) $\dfrac{\sin x}{\cos x} = \dfrac{1}{2}$; $\tan x = 0.5$

$\dfrac{\sin x}{\cos x} = \tan x$ *used to get* $\tan x = k$

or identity $\cos^2 x + \sin^2 x = 1$ *used to get either* $\sin^2 x = p$ *or to get* $\cos^2 x = q$, *(p and q must be between 0 and 1)*

M1

$\tan x = 0.5$

Either $\tan x = \frac{1}{2}$ or $\cos x = \pm\sqrt{\frac{4}{5}}$ *(= ±0.894..)*

or $\sin x = \pm\sqrt{\frac{1}{5}}$ *(= ±0.447..)*

A1

$x = \alpha$ or $\pi + \alpha$ where $\alpha = \tan^{-1}(k)$

Correct method to find 2nd angle. Any in wrong ft quadrants then m0. In case of squaring method candidates must also have rejected the extra 'quadrants' for the m1. Condone degrees or mixture

m1

x = 0.464, 3.61

Both. Condone > 3sf [0.463(6..), 3.60(5..or 6.)]

Accept pair of truncated values [0.463, 3.60]

Ignore any answers outside interval 0 to 6.28

A1

4

[10]

Q28.

(a) (i) ($\sin^{-1} \pm 0.25 =$) ± 14.5

PI by sight of 194.5 etc

condone ±14.4

M1

θ = 194.5, 345.5 (AWRT)

no extras in interval, ignore answers outside interval

A1

2

(ii) $2\cot^2(2x + 30) = 2 - 7\operatorname{cosec}(2x + 30)$

condone replacing 2x + 30 by Y

$2(\operatorname{cosec}^2(2x + 30) - 1) = 2 - 7\operatorname{cosec}(2x + 30)$

correct use of $\operatorname{cosec}^2 Y = 1 + \cot^2 Y$

M1

$2\operatorname{cosec}^2(2x + 30) + 7\operatorname{cosec}(2x + 30) - 4 (= 0)$

must be in this form

A1

$(2 \operatorname{cosec} (2x + 30) \pm 1)(\operatorname{cosec} (2x + 30) \pm 4)$ (= 0)

attempt at factorisation

m1

$\operatorname{cosec}(2x + 30) = \frac{1}{2}$ or −4

must be this line using f (2x + 30)

A1

$2x + 30 = 194.5, 345.5$

$x = 82.2, 157.8$ (AWRT)

one correct answer, allow 82.3, ignore extra solutions

B1

CAO both answers correct and no extras in interval, ignore answers outside interval

B1 **6**

(b) stretch (I)

scale factor $\frac{1}{2}$ (II)

parallel to x–axis (III)

I and either II or III

M1

I + II + III

A1

translate

E1

$\begin{pmatrix} -15 \\ 0 \end{pmatrix}$

condone '15 to left' or '–15 in x (direction)'

B1 **4**

alternative method

translate

(E1)

$\begin{pmatrix} -30 \\ 0 \end{pmatrix}$

(B1)

stretch

as above

(M1)

scale factor $\frac{1}{2}$

parallel to x–axis

as above

(A1)

[12]

Q29.

(a) $x^2 - 8x + 15 + 2$

Terms in x must be collected, PI

B1

their $x - 4^2$ $(+ k)$

ft $x - p^2$ for their quadratic

M1

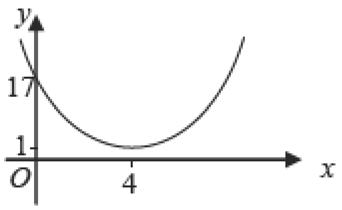
$= x - 4^2 + 1$

ISW for stating if $p = -4$ correct expression seen

A1

3

(b) (i)



$\cup$ *shape in any quadrant (generous)*

M1

correct with min at (4, 1) stated or 4 and 1 marked on axes
condone within first quadrant only

A1

crosses y-axis at (0, 17) stated or 17 marked on y-axis

B1

3

(ii) $y = k$

y = constant

M1

$y = 1$

Condone $y = 0x + 1$

A1

2

(c) Translation (*not shift, move etc*)

and no other transformation

E1

with vector $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

One component correct
or ft either their p or q

M1

CSO; condone 4 across, 1 up; or two separate vectors etc

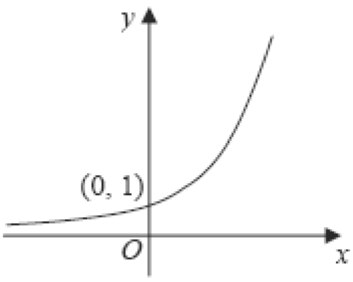
A1

3

[11]

Q30.

(a)



Shape with some indication of asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis at (0, 1) stated/indicated … (accept 1 on y-axis as equivalent)

B1

2

(b) (i) $h = 0.5$

PI

B1

$f(x) = 2^x$

$I \approx h/2\{\ldots\}$

$\{\ldots\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$

OE summing of areas of the 4 'trapezia'

M1

$\{\ldots\} = 1 + 4 + 2(\sqrt{2} + 2 + \sqrt{8})$

$= 5 + 2 \times 6.2426\ldots = 17.485\ldots$

OE Accept 2 dp (rounded or truncated) as evidence for surds

A1

$(I \approx)$ 4.3713… = 4.37 (to 3 sf)

CAO Must be 4.37

***SC** for those who use 5 strips, max possible is B0M1A1A0*

A1

4

(ii) Increase the number of ordinates

OE

E1

1

(c) Translation;

Accept 'translat…' as equivalent
[T or Tr is NOT sufficient]

B1;

$\begin{bmatrix} -7 \\ 3 \end{bmatrix}$

B1 for each component of the vector.
Condone if the equiv 2 vectors are given.
*Accept **full** equivalent to vector(s) in words provided linked to 'translation/ move/shift' and **correct** directions.*
*(No marks if **different** transformations)*

B1;B1

3

(d) $8 = 2^k + 3 \Rightarrow 2^k = 5$

Correct subst. and an attempted rearrangement to $2^k = N$. PI by $k = \frac{\log 5}{\log 2}$

M1

$k = \log_2 5$

Accept m = 2, n = 5

A1

2

[12]

Q31.

(a) (y =) 1

B1

1

(b) $h = 0.2$

PI

B1

$f(x) = 2^{4x}$
$I \approx h/2 \{\ldots\}$
$\{.\} = f(0) + f(1) + 2[f(0.2) + f(0.4) + f(0.6) + f(0.8)]$

OE summing of areas of the 'trapezia'..

M1

$\{.\} = 1 + 16 + 2(2^{0.8} + 2^{1.6} + 2^{2.4} + 2^{3.2})$
$= 1 + 16 + 2(1.741.. + 3.031.. + 5.278.. + 9.1895..)$
$= [17 + 2 \times 19.24...]$

OE Accept 2dp rounded or truncated evidence

A1

I = 5.55 (to 2 dp)

Must be 5.55

A1

4

(c) Stretch(**I**) in y-direction(**II**) scale factor $\frac{1}{8}$ (**III**)

*Need (**I**) **and** either (**II**) or (**III**)*

M1

*Need (**I**) and (**II**) and (**III**)*

A1

ALTn: Translation with an indication that the translation is in the x-direction (B1)

$\begin{bmatrix} \frac{3}{4} \\ 0 \end{bmatrix}$ (B1)

*Combination of **different** transformations scores 0/2*

2

(d) $g(x) = 2^{4(x-1)} - \frac{1}{2}$

B1 for either $2^{4(x+1)} - \frac{1}{2}$ or for $2^{4(x-1)} + \frac{1}{2}$ or for $2^{4x-1} - \frac{1}{2}$

B2,1,0

At Q, $y = 0 \Rightarrow 2^{4(x-1)} = 2^{-1}$

Reaches a stage from which linear eqn can be stated directly eg an alternative stage is $4(x-1)\log 2 = -\log 2$

M1

$\Rightarrow$ $4x - 4 = -1$ $\Rightarrow$ $x = 0.75$

NMS mark as 4 or 0

A1

4

(e) (i) $\log_a k = \log_a 2^3 + \log_a 5 - \log_a 4$

$\log_a k = \log_a (2^3 \times 5) - \log_a 4$

One law of logs used

M1

A second law of logs used; could be

$\log_a k = \log_a 2^3 + \log_a \left(\frac{5}{4}\right)$

M1

$\log_a k = \log_a \left(\frac{2^3 \times 5}{4}\right) = \log_a 10 \Rightarrow k = 1$

CSO AG

A1

3

(ii) $2^{4x-3} = \frac{5}{4}$ so

*Equate y's, take logs (to any base) of both sides **and** apply 3rd law of logs.*

$(4x - 3) \log_{10} 2 = \log_{10} \frac{5}{4}$

M1

$$x = \frac{3\log_{10} 2 + \log_{10}\left(\frac{5}{4}\right)}{4\log_{10} 2}$$

m1

$$x = \frac{\log_{10} 10}{4\log_{10} 2} \text{ so } x = \frac{1}{4\log_{10} 2}$$

A1

Altn $4x \log 2 = \log \left(\frac{5}{4} \times 2^3\right)$

Rearrange correctly to $x = \ldots\ldots$

Altn $4x \log 2 = \log 10$

In both cases, log term(s) must have same base and expressions must be in an exact form, ie not approx. dec. vals

CSO AG Must be clear evidence that base 10 is used, also be convinced

3

[17]

Q32.

(a) $y = e^x \to e^{2x} - 1$

Stretch (I)

scale factor $\frac{1}{2}$ (II)

I + (II or III)

M1

in x-direction (III)

I + II + III

A1

Translation

Allow "translate"

E1

$\begin{bmatrix} 0 \\ -1 \end{bmatrix}$

OE "1 unit down" etc

B1

4

(b) $x = 0$ $y = 6$ or $(0, 6)$

Both coordinates must be stated, not simply 6 marked on diagram

B1

1

(c) (i) $e^{2x} - 1 = 4e^{-2x} + 2$

$e^{4x} - e^{2x} = 4 + 2e^{2x}$

or $(e^{2x})^2 - e^{2x} = 4 + 2e^{2x}$

Multiplying both sides by e^{2x}

M1

$(e^{2x})^2 - 3e^{2x} - 4 = 0$

AG With no errors seen

A1

2

(ii) $(e^{2x} - 4)(e^{2x} + 1)$

$(e^{2x} \pm 4)(e^{2x} \pm 1)$

M1

$x = \ln 2$ or $\frac{1}{2} \ln 4$

A1

Reject $e^{2x} = -1$ OE

Eg $e^{2x} > 0$, $e^{2x} \neq -1$, impossible etc

A1

3

(d) $\int (4e^{-2x} + 2)\,dx$ (I)

$$= \left[\frac{4e^{-2x}}{-2} + 2x\right]_0^{\ln 2}$$

I or II attempted and e^{-2x} or e^{2x} integrated correctly

M1

$$= \left(\frac{4e^{-2\ln 2}}{-2} + 2\ln 2\right) - \left(\frac{4}{-2} + 0\right)$$

F ['their ln 2' from (c)(ii)] – F[0]

m1

$$= -\frac{1}{2} + 2\ln 2 + +2 = \frac{3}{2} + 2\ln 2$$

$\int (e^{2x} - 1)\,dx$ (II)

$$= \left[\frac{e^{2x}}{2} - x\right]_0^{\ln 2}$$

Both I and II correctly integrated

A1

$$= \left(\frac{e^{2\ln 2}}{2} - \ln 2\right) - \left(\frac{1}{2} - 0\right)$$

$$= 2 - \ln 2 - \frac{1}{2} = \frac{3}{2} - \ln 2$$

$$A = \left(\frac{3}{2} + 2\ln 2\right) - \left(\frac{3}{2} - \ln 2\right)$$

Attempt to find difference of 'their I – their II'

B1ft

$$= 3\ln 2 \quad \text{or} \quad \ln 8 \quad \text{or} \quad \frac{3}{2}\ln 4 \quad \text{OE}$$

CSO must be exact

A1

Alternative

$$A = \int \left(4e^{-2x} + 2\right)dx - \int (e^{2x} - 1)\ dx$$

Condone functions reversed

(B1)

$$= \int_{(0)}^{(\ln 2)} (4e^{-2x} - e^{2x} + 3)\,dx$$

$$= \left[\frac{4e^{-2x}}{-2} - \frac{e^{2x}}{2} + 3x\right]_0^{\ln 2}$$

e^{2x} or e^{-2x} correctly integrated

(M1)(A1)

$$= \left(-2e^{-2\ln 2} - \frac{1}{2}e^{2\ln 2} + 3\ln 2\right) - \left(-2 - \frac{1}{2}\right)$$

Correct substitution of their ln 2 from (c)(ii) into their integrated expression

(m1)

$$= 3\ln 2 \quad \text{or} \quad \ln 8 \quad \text{or} \quad \frac{3}{2}\ln 4 \quad \text{OE}$$

CSO must be exact

(A1)

5

[15]

Q33.

(a) (i) $(x + 1)^2$

$p = 1$

B1

$+ 4$

$q = 4$

B1

2

(ii) $(x + 1)^2 \geq 0 \Rightarrow (x + 1)^2 + 4 > 0$

$(\Rightarrow x^2 + 2x + 5 > 0$ for all values of $x)$

Condone if they say $(x + 1)^2$ positive and adding 4 so always positive

E1

1

(b) (i) $x = -1$ **or** $y = 4$

ft their $x = -p$ or $y = q$

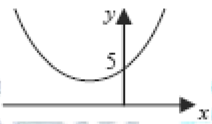
M1

Minimum point is $(-1, 4)$

A1

2

(ii)



Sketch roughly as shown

B1

y-intercept 5 or (0, 5) marked or stated

B1

2

(c) Translation (not shift, move etc)

and NO other transformation stated

E1

through $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ (or 1 left, 4 up etc)

either component correct or ft their $-p$, q

M1

correct translation
M1, A1 independent of E mark

A1

3

[10]

Q34.

(a) (i) $\frac{dy}{dx} = 3x^{\frac{1}{2}}$

$kx^{\frac{1}{2}}$ *with or without + c*

M1

$= 6$ {when $x = 4$}

Must be 6 and seen in (a)(i)
6 + c is A0

A1cao

2

(ii) y-coordinate of A = $2 \times 4^{\frac{3}{2}} (= 16)$

Substitute $x = 4$ in $y = 2x^{\frac{3}{2}}$

M1

$6 \times m' = -1$

$m_1 \times m_2 = -1$ OE used with c's value of
$\frac{dy}{dx}$ *when $x = 4$. PI*

M1

$y - 16 = m(x - 4)$

dep on 1st M1 in (a)(ii)
m must be numerical

m1

$y - 16 = -\frac{1}{6}(x - 4)$

ACF

A1

4

(b) (i) $\int 8x^{\frac{1}{2}}\,dx = \frac{8}{3/2}x^{\frac{1}{2}+1}\{+c\}$

Index raised by 1

M1

$= \frac{16}{3}x^{\frac{3}{2}}\{+c\}$

Condone missing '+ c'
Coefficient must be simplified

A1

2

(ii) $\int 2x^{\frac{3}{2}}\,dx = \frac{2}{5/2}x^{\frac{5}{2}}\{+c\} \quad \{= \frac{4}{5}x^{\frac{5}{2}}\{+c\}\}$

Can award for unsimplified form

B1

$\int_0^4 8x^{\frac{1}{2}}\,dx - \int_0^4 2x^{\frac{3}{2}}\,dx$

Ignore limits here

M1

$= \frac{16}{3}(4)^{\frac{3}{2}} - 0 - \left[\frac{4}{5}(4)^{\frac{5}{2}} - 0\right]$

F(4) – F(0) used in either; {F(0) = 0 PI}
Cand. must be using F(x) as a result of his/her integration in (b)(i) or in the (b)(ii) B1 line above

M1

$= \frac{256}{15}$

Accept any value from 17.04 to 17.1 inclusive in place of 256/15

A1

4

(c) Translation

Accept 'translat…' as equivalent

[T or Tr is NOT sufficient]

B1

$\begin{bmatrix} -3 \\ 0 \end{bmatrix}$

Accept equivalent in words provided linked to 'translation/move/shift' (B0B0 if >1 transformation)

B1

2

[14]

Q35.

(a) $P(-1, \pi)$

Condone (–1, 180°)

B1

Q (1, 0)

B1

2

(b) Translate

E1

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

or equivalent in words

B1

Stretch SF 2 // y-axis

Stretch + one other correct

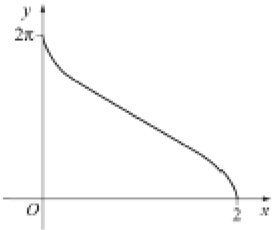
M1

all correct

A1

4

(c)



Correct shape in 1st quadrant

B1

2π and 2 marked correctly

B1

2

(d) (i) $\frac{y}{2} = \cos^{-1}(x-1)$

M1

$\cos\left(\frac{y}{2}\right) = x - 1$

$x = \cos\left(\frac{y}{2}\right) + 1$

A1

2

(ii) $-\frac{1}{2}\sin\left(\frac{y}{2}\right)$

k sin (...)

M1

$\frac{dx}{dy}$ *correct*

A1

At $y = 2$, $\left(\frac{dx}{dy} =\right) - \frac{1}{2}\sin 1$

Condone AWRT –0.42

A1

3

[13]